

# **NEWT'S AND RAPH'S**

Submitted by:

Ladores, Oliver P.

Galvez, Khee Jay

Billena, Dhominick John

Submitted on:

December 15, 2025

## Contents

|                        |    |
|------------------------|----|
| Project Overview ..... | 3  |
| Objectives .....       | 4  |
| Showcase .....         | 5  |
| Assets .....           | 8  |
| Recommendations .....  | 9  |
| Citations .....        | 10 |

## Project Overview

This project aims to implement the Newton-Raphson Method for Optimization [1] polynomial function  $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$ , in matrix form:

$$P_{f(x)} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \end{bmatrix}$$

It does so by multiplying it with a matrix constructed from the size of the polynomial in matrix form, the matrix for the derivative operation constructed being:

$$D_{\text{matrix}} = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots \\ & 0 & 2 & 0 & \dots \\ & & 0 & 3 & \dots \\ & & & 0 & \ddots \end{bmatrix}$$

Figure 1: The derivative operation for polynomials, represented as a derivative.

By multiplying them both, we get  $f'(x)$  [2]:

$$P_{f'(x)} = D_{\text{matrix}} \times P_{f(x)}$$

Evaluating the polynomial at a given point is simple, by raising  $x$  elementwise to the index of the polynomial in matrix form (the degree of each of the terms), then multiplying it to the coefficient defined in the polynomial matrix form.

$$f(x) = \sum_{i=0}^n x^{\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \\ n \end{bmatrix}} \circ \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$$

$$\text{where } x^{\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \\ n \end{bmatrix}}$$

is the elementwise operation of raising  $x$  to every element from 1 to  $n$

## Objectives

- Develop a library that can calculate the Newton-Raphson method and other prerequisites operations (differentiation and polynomial evaluation).
- Provide a simple user interface that can expose the functionality of the library to a non-command line user.
- Check if the library correctly calculates the optimal values of a given function, up to tolerance.

## Showcase

# Newton-Raphson Method for Optimization

Add or delete additional terms:

Add

Delete

count = 4

$c$

-3

$x$

-5

$x^2$

-7

$x^3$

2

$x^4$

4

Submit

Figure 2: The title bar, plus the inputs for each of the terms of the polynomial.

## Polynomial

$$f(x) = -3.0 - 5.0x - 7.0x^2 + 2.0x^3 + 4.0x^4$$

## Derivative

$$f'(x) = -5.0 - 14.0x + 6.0x^2 + 16.0x^3$$

## Second Derivative

$$f''(x) = -14.0 + 12.0x + 48.0x^2$$

Figure 3: Show the input function, typeset with LaTeX.

|               |
|---------------|
| 0             |
| Initial Guess |
| 0.05          |
| Tolerance     |

Figure 4: Enter initial guess and tolerance.

## Newton-Raphson Method for Optimization

Polynomial:  $f(x) = -3.0 - 5.0x - 7.0x^2 + 2.0x^3 + 4.0x^4$

Initial Guess:  $x_1 = 0.0$

Tolerance:  $\epsilon = 0.05$

Optimal value:  $-0.35714285714285715$

Optimal solution:  $-2.133173677634319$

Figure 5: Display results of Newton-Raphson method for optimization



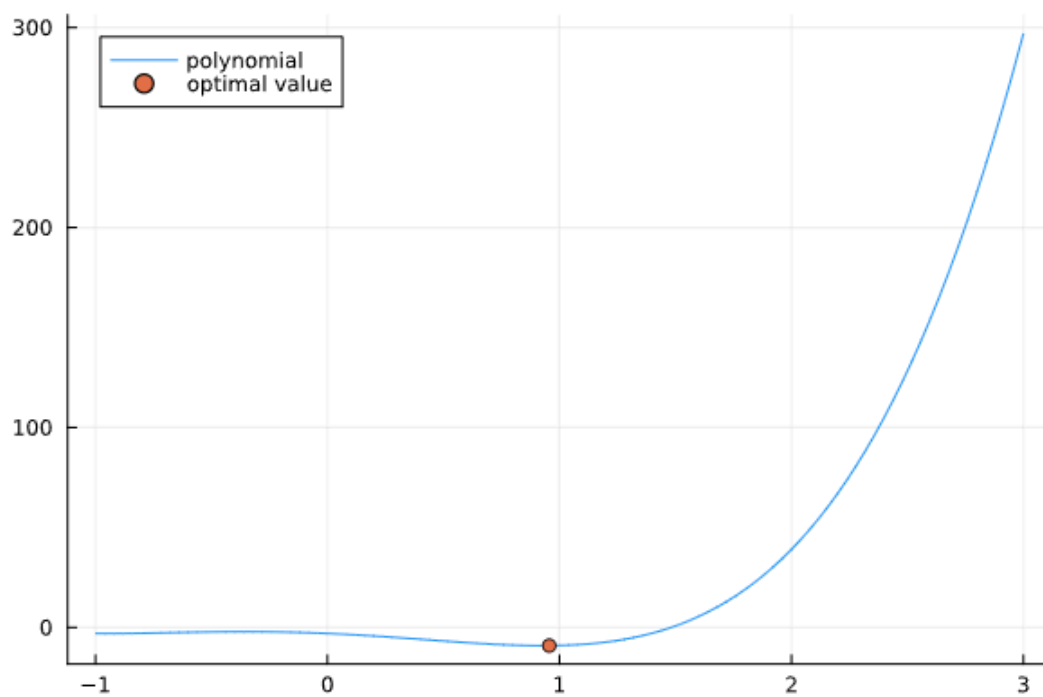
Display table of optimization



Plot optimal values

Min

Max



4x2 DataFrame

| Row | x_i<br>Float64 | ε<br>Bool |
|-----|----------------|-----------|
| 1   | 2.03304        | false     |
| 2   | 1.43059        | false     |
| 3   | 1.09435        | false     |
| 4   | 0.955976       | true      |

Figure 6: Display graph and table.

## Assets

- [GitHub Link](#)

To run, it needs the Julia programming language, and the Pluto Julia library.

- [UI only \(code will \*\*not\*\* run\)](#)



## Recommendations

- Extend the solver to be able to calculate derivatives for trigonometric and exponential functions.
- Implement an os-native user interface.

## Citations

- [1] "Newton's method in optimization." Accessed: Dec. 15, 2025. [Online]. Available: [https://en.wikipedia.org/w/index.php?title=Newton%27s\\_method\\_in\\_optimization&oldid=1311976880](https://en.wikipedia.org/w/index.php?title=Newton%27s_method_in_optimization&oldid=1311976880)
- [2] "Derivatives via matrix multiplication." [Online]. Available: <https://youtu.be/SSPbgRQCqUo>